From last time

∴ Not geom quot.

(Problem: 0 lies in every orbit closure, wants to get rid of it)

Rmk (4.34) • Used geom. reductive to show $\mathcal{O}(X)^G$ f.g. (Nagata)
(Hilbert, Mumford: used lin. red.)
• Used lin. red. to show Lemma 4.24,
which implies $X \to X/\!/G$ surj.
• Used geom. reductive to show Lemma 4.29
(some separation statement)

Section 4.6 Setting: G reductive, X affine, $G \curvearrowright X$.

Def (4.35) $x \in X$ is stable if • its orbit is closed in X, and
• $\dim G_x = 0$ (equiv, $\dim G \cdot x = \dim G$)
Stabilizer

Let X^s denote the set of stable points,

Prop 4.36 Let $G \curvearrowright X$, $Y = X/\!/G$.
Then ① $X^s \subseteq X$ open G-inv.
② $Y^s := \varphi(X^s) \subseteq Y$ open and $X^s = \varphi^{-1}(Y_s)$
③ $\varphi : X^s \to Y^s$ geom quotient.

Proof: ① G-inv: Clear: $\forall g \in G$, $\begin{cases} G\cdot x \text{ closed} \Leftrightarrow G\cdot(g\cdot x) \text{ closed} \\ \dim G_x = 0 \Leftrightarrow \dim G_{g\cdot x} = 0 \end{cases}$

$X_+ := \{x \in X : \dim G_x > 0\}$ closed by semi-continuity. (Lemma 3.21)

Let $x \in X^s$.

Lemma 4.29 $\Rightarrow \exists f \in \mathcal{O}(X)^G$ s.t.

$f(X_+) = 0$, $f(G\cdot x) = 1$

(note: elements in $G\cdot x$ has 0-dim stabilizer $\Rightarrow G\cdot x \cap X_+ = \emptyset$)

Claim: $X_f \subseteq X^s$

Claim $\Rightarrow x \in X_f$.

$\therefore X^s$ open

Pf of claim: Suffices to show elements in X_f has closed orbits.

Suppose $z \in X_f(k)$ has a non-closed orbit.

Can take $w \in \overline{G\cdot z} - G\cdot z$ on the boundary.

$f(z) \neq 0 \Rightarrow f(g\cdot z) \neq 0 \ \forall g \Rightarrow f(\overline{G\cdot z})$ avoids 0.

$\therefore w \in X_f(k) \Rightarrow \dim G_w = 0$.

But $\dim G\cdot w < \dim G\cdot z \Rightarrow \dim G_w > \dim G_z = 0$

contradiction.

② $\varphi(X_f) = Y_f \ \forall f \in \mathcal{O}(X)$ & $X_f = \varphi^{-1}(Y_f)$

($X^s \subseteq X$ open ?) $\overset{(?)}{\Rightarrow} Y^s \subseteq Y$ open & $X_f = \varphi^{-1}(Y_f)$

(3) $G \curvearrowright X^s$ is closed (i.e. $G \cdot x$ closed for all x)

$\therefore$ Cor 3.32 $\Rightarrow$ $\varphi : X^s \to Y^s$ is geom. quot.

Eg (4.37) $\mathbb{G}_m \curvearrowright \mathbb{A}^2$

$t \cdot (x,y) = (tx, t^{-1}y)$

Closed orbits: $\{xy = \alpha\}$ $\dim G_x = 0$

$\{0\}$ $\dim G_x = 1$

$\therefore X^s = \{(x,y) \in \mathbb{A}^2 : xy \neq 0\}$ ↓ need to throw away

Eg. (4.38) $\mathbb{G}_m \curvearrowright \mathbb{A}^n$

$t \cdot (x_1, \ldots, x_n) = (tx_1, \ldots, tx_n)$

Closed orbit: $\{0\}$ $\dim G_x = 1$

$\therefore X^s = \emptyset$

Eg (4.39) $GL_2 \curvearrowright M_{2\times 2} = \operatorname{Spec} k[x_{11}, x_{12}, x_{21}, x_{22}]$

$g \cdot A = gAg^{-1}$

Orbits: $A = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}$ w/ $\alpha \neq \beta$

closed orbit

$G_A = \begin{pmatrix} * & \\ & * \end{pmatrix}$

$\dim G_A = 2$

- $A = \begin{pmatrix} \alpha & 1 \\ 0 & \alpha \end{pmatrix}$

NOT closed orbit:

$$\lim_{t \to 0} \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix} \begin{pmatrix} t^{-1} & \\ & t \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}$$

$G_A = \langle A \rangle$

$\dim G_A = 2$

- $A = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}$

closed orbit

$G_A = GL_2$

$\dim G_A = 4$

Fact: $\mathcal{O}(M_{2\times 2})^{GL_2} = k[\mathrm{tr}, \det]$

(Gives a GIT quot.)

Note: $G_m I_2$ fixes every point

$\therefore \dim G_A \geq 1$

$\therefore X^s = \emptyset$.

Rmk (4.41) Since $G_m I_2$ acts trivially, causing stabilizers

Better to act by $GL_2/\mathbb{G}_m I_2 = PGL_2$ instead.
(Lecture notes: Use SL_2)

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Exercise sheet 6

Ex 1. Setting: $R = k[x_1, x_2]$, $\Gamma \subseteq SL_2$ finite, specified below.
Then $\Gamma \curvearrowright R$, want to find R^Γ.

(a) Fix $n \in \mathbb{Z}_{>0}$ s.t. $\mathrm{char}(k) \nmid n$

$$\Gamma = \left\{ \begin{pmatrix} \eta & 0 \\ 0 & \eta^{-1} \end{pmatrix} : \eta^n = 1 \right\}$$

Note: Since $\mathrm{char}(k) \nmid n$, $x^n - 1$ is separable.
$\therefore \#\{\eta : \eta^n = 1\} = n$

Calculation: Fix ζ a primitive n-th root of unity.
Let $g = \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}$, then $\Gamma = \langle g \rangle$.
Then $f \in R^\Gamma \Leftrightarrow g \cdot f = f$.

Suppose $f \in R^\Gamma$. let $f = \sum_{\underline{a} \in \mathbb{Z}_{\geq 0}^2} c_{\underline{a}} x^{\underline{a}}$.

Then $g \cdot f = \sum_{\underline{a} \in \mathbb{Z}_{\geq 0}^2} c_{\underline{a}} \zeta^{-a_1} x_1^{a_1} \zeta^{a_2} x_2^{a_2}$.

$g \cdot f = f \Rightarrow \quad c_{\underline{a}} = c_{\underline{a}} \zeta^{a_2 - a_1}$

$\Rightarrow \quad c_{\underline{a}} = 0 \quad$ or $\quad \zeta^{a_2 - a_1} = 1$

$\therefore$ If $a_2 - a_1 \not\equiv 0 \pmod n$, then $c_{\underline{a}} = 0$

$\therefore \quad R^\Gamma = \langle x_1^i x_2^j : i - j \equiv 0 \pmod n \rangle$ (linear span)

$= k[x_1 x_2, x_1^n, x_2^n]$ (algebra generators)

$\cong k[u_1, u_2, u_3] / (u_2 u_3 - u_1^n)$

$f \in R^\Gamma$ must be sum of monomials in the for $x_1^i x_2^j : i - j \equiv 0 \ (n)$

Not smooth at $(0,0,0)$

(b) Fix $n \in \mathbb{Z}_{>0}$ s.t. $\mathrm{char}(k) \nmid 2n$

$$\Gamma = \left\{ \begin{pmatrix} \eta & 0 \\ 0 & \eta^{-1} \end{pmatrix}, \begin{pmatrix} 0 & \eta \\ -\eta^{-1} & 0 \end{pmatrix} : \eta^{2n} = 1 \right\}$$

Note: For odd k, $\left\{ \begin{pmatrix} \eta & 0 \\ 0 & \eta^{-1} \end{pmatrix}, \begin{pmatrix} -\eta & 0 \\ 0 & -\eta^{-1} \end{pmatrix} : \eta^k = 1 \right\}$ is NOT a group

Calculation: Fix ζ a primitive $2n$-th root of unity.

Let $g = \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}$, $h = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, then $\Gamma = \langle g, h \rangle$.

↑ Note: This is the dicyclic group: $\langle g, h : g^{2n} = 1, h^2 = g^n, h^{-1} g h = g^{-1} \rangle$

Then $f \in R^\Gamma \Leftrightarrow \begin{cases} g \cdot f = f \\ h \cdot f = f \end{cases}$

Suppose $f \in R^\Gamma$, let $f = \sum_{\underline{a} \in \mathbb{Z}^2_{\geq 0}} c_{\underline{a}} x^{\underline{a}}$.

Then $g \cdot f = \sum_{\underline{a} \in \mathbb{Z}^2_{\geq 0}} c_{\underline{a}} \zeta^{-a_1} x_1^{a_1} \zeta^{a_2} x_2^{a_2}$.

$\therefore$ If $a_2 - a_1 \not\equiv 0 \pmod{2n}$, then $c_{\underline{a}} = 0$.

$h \cdot f = \sum_{\underline{a} \in \mathbb{Z}^2_{\geq 0}} c_{\underline{a}} (-x_2)^{a_1} (x_1)^{a_2}$

$\therefore \quad C_{(a_1,a_2)} = (-1)^{a_1} C_{(a_2,a_1)}$

In particular: If $a_1 = a_2$, $C_{(a,a)} = (-1)^a C_{(a,a)} \Rightarrow$ For odd a, $C_{(a,a)} = 0$.

$\therefore \quad R^\Gamma = \langle x_1^{2i} x_2^{2i} \rangle + \langle x_1^i x_2^j + (-1)^i x_1^j x_2^i : i \equiv j \pmod{2n} \rangle$ (linear span)

$= k[x_1^2 x_2^2, \; x_1^{2n} + x_2^{2n}, \; x_1 x_2 (x_1^{2n} - x_2^{2n})]$ (algebra generators)

$\cong k[u_1, u_2, u_3] / (4u_1^{n+1} - u_1 u_2^2 + u_3^2)$

Not smooth at $(0,0,0)$

Ex 3. Let $G = G_a$. Want $O(V)^{G_a}$ for the following representations:

(a) $G \curvearrowright \mathbb{A}^3$ by $\rho(c) := \begin{pmatrix} 1 & 2c & c^2 \\ & 1 & c \\ & & 1 \end{pmatrix}$

Calculation: $c \cdot x_1^{a_1} x_2^{a_2} x_3^{a_3} = x_1^{a_1} (x_2 - 2cx_1)^{a_2} (x_3 - cx_2 - c^2 x_1)^{a_3}$

(b) $G \subset \mathbb{A}^4$ by $P(c) := \begin{pmatrix} 1 & c & & \\ & 1 & & \\ & & 1 & c \\ & & & 1 \end{pmatrix}$

Calculation: $c \cdot x_1^{a_1} x_2^{a_2} x_3^{a_3} x_4^{a_4} = x_1^{a_1} (x_2 - cx_1)^{a_2} x_3^{a_3} (x_4 - cx_3)^{a_4}$

Suppose $f \in \mathcal{O}(V)^G$:

Action on deg-2 forms:

$x_1^2 \mapsto x_1^2$

$x_1x_2 \mapsto x_1x_2 - cx_1^2 \Rightarrow [x_1x_2]f = 0$

$x_1x_3 \mapsto x_1x_3$

$x_1x_4 \mapsto -cx_1x_3 + x_1x_4$

$x_2^2 \mapsto c^2x_1^2 - 2cx_1x_2 + x_2^2 \Rightarrow [x_1^2]f = 0$

$\Rightarrow [x_1x_4]f = [x_2x_3]f$

$x_2x_3 \mapsto -cx_1x_3 + x_2x_3$

$x_2x_4 \mapsto c^2x_1x_3 - cx_1x_4 - cx_2x_3 + x_2x_4 \Rightarrow [x_2x_4]f = 0$

$x_3^2 \mapsto x_3^2$

$x_3x_4 \mapsto -cx_3^2 + x_3x_4 \Rightarrow [x_3x_4]f = 0$

$x_4^2 \mapsto c^2x_3^2 - 2cx_3x_4 + x_4^2 \Rightarrow [x_4^2]f = 0$

(Reasoning unknown)

$\Rightarrow \mathcal{O}(V)^G = k[x_1, x_3, x_1x_4 - x_2x_3]$

Peter's sharing:

Klein (late 1800's): Finite subgroups of $SL_2(\mathbb{C})$ up to conjugacy are one of following:

- Type A_n: $\mathbb{Z}/n+1 = \left\langle \begin{pmatrix} e^{2\pi i/n} & 0 \\ 0 & e^{-2\pi i/n} \end{pmatrix} \right\rangle$

  D_n: $\left\langle \begin{pmatrix} e^{2\pi i/n} & 0 \\ 0 & e^{-2\pi i/n} \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\rangle$

  E_6 rotational symmetry of tetrahedron

  E_7 octahedron

  E_8 icosahedron

Also showed they have 3 generating invariant poly.
$(\mathbb{C}[x,y]^G \cong \mathbb{C}[p_1, p_2, p_3])$

Du Val (1930's): Studied quotients

$$\mathbb{C}^2/G \cong \operatorname{Spec} \mathbb{C}[x,y]^G \hookrightarrow \mathbb{C}^3$$

isolated singularity at origin

Resolution graph: $Y \xrightarrow{\pi} \mathbb{C}^2/G$ (only singular at 0)

$\uparrow$ minimal resolution of singularity

nodes: $\mathbb{P}^1$'s in $\pi^{-1}(0)$ (Blowups at 0 give $\mathbb{P}^1$'s)

edge between if $\mathbb{P}^1$'s intersect

A_n :

D_n :



E_6 :



E_7 :



E_8 :



McKay (1980's):

Noticed: (1) # of non-trivial irreducible reps of G are in bijection w/ # of $\mathbb{P}^1$'s in minimal res.

(2) Can also build the same Dynkin diagram from a rep. theory construction

Auslander (?): $\mathbb{C}[x,y]^G \hookrightarrow \mathbb{C}[x,y]$

Write $\mathbb{C}[x,y]$ as a $\mathbb{C}[x,y]^G$-mod, summands correspond to Dynkin diagram nodes.

What about $G \subseteq GL_2(\mathbb{C})$? (reflection: fixing hyperplanes)

Two cases: ① G is "small" if no reflections

② G gen. by reflections

- Chevalley-Shephard-Todd (1950's):

$\mathbb{C}^2/G \cong \mathbb{C}^2$

E.g. $G = \left\langle \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

$$\mathbb{C}[x,y]^G = \mathbb{C}[x^2, y^2]$$